

ATLAS STRUCTURAL SYSTEMS LLC

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BID PROPOSAL - LUXURY MIXED-USE HIGH-RISE FOUNDATION & PODIUM PACKAGE

Date: September 20, 2026

Project: Lumina Hotel & Residences – 42-Story Mixed-Use Tower (Four Seasons Operator) – Foundation & Podium Transfer Package

Owner: BlueRock Institutional Partners LP (Blackstone JV) – c/o BrightPath Property Management

Location: 1800 West Loop South, Houston, TX 77027

Proposal ID: ATL-2026-FND-091A | Valid 60 Days

Scope: Sub-scope: Drilled piers (12 ea), Grade beams, Post-tensioned podium slab (Level 2 transfer), Waterproofing & Drainage Board, Limited Dewatering



PAGE 1: EXECUTIVE SUMMARY & PROJECT UNDERSTANDING

Atlas Structural Systems proposes to deliver the critical path foundation and podium transfer structure for the Lumina Hotel & Residences, a 42-story luxury tower (198 hotel keys, 140 luxury condos, 18k sqft retail). This \$98,750 bid is for Phase 1A only – substructure below Level 2 podium. The site sits on expansive Gulf Coast clay over Beaumont clay with water table at -12 ft. Our solution: 36-inch diameter drilled piers belled to 60-inch at 38-42 ft depth to competent bearing stratum (2000 psf), with 30-inch x 36-inch grade beams and 12-inch post-tensioned podium slab (PT design per PTI DC45). Differentiator: Atlas uses self-consolidating concrete (SCC) mix with 15% fly ash for reduced heat of hydration, and includes 3D BIM coordination for MEP penetrations in podium (often missed). Includes structural engineer of record (SER) review allowance.

PAGE 2: STRUCTURAL ENGINEERING & MATERIALS

Concrete: 5000 psi @ 28 days for piers (ASTM C150 Type I/II), 6000 psi for podium slab PT, w/c ratio 0.42 max, air entrainment 5-7%, slump 8-10 inch SCC. Mix design by CTS Houston lab, submittals included.

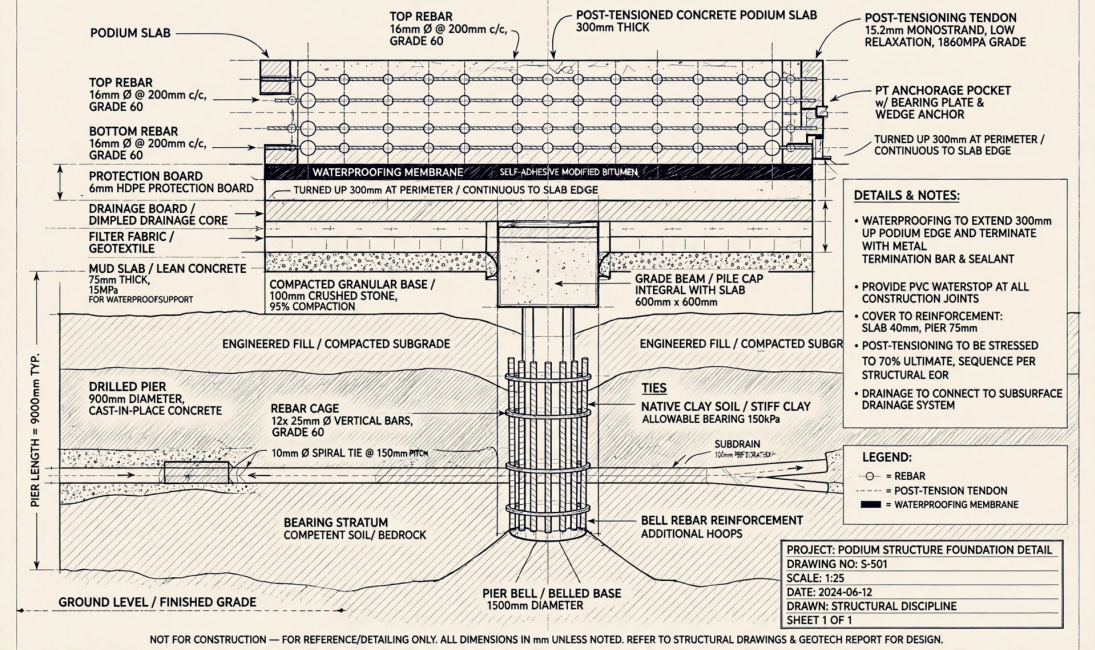
Rebar: Grade 60, #6 @ 12-inch c/c top/bottom podium, #8 vertical in piers (12 bars per pier), #4 ties @ 12-inch. Epoxy coated in splash zone. Lap splices per ACI 318-19.

Post-Tensioning: 0.5-inch 270K low-relaxation unbonded tendons, 33 ksi effective after losses, 70% ultimate stressing sequence per SER. Two-stage stressing (50% at 75% f'c, 100% at 100%). Pocket formers & grout caps included.

Waterproofing: Below podium: Henry Blueskin WP 200 self-adhesive membrane, 6mm HDPE protection board, dimple drainage core with filter fabric to perimeter subdrain (4-inch perf PVC to sump).

Foundation Detail: See engineering detail – belled pier with 1500mm bell, engineered fill 95% compaction, mud slab 3-inch lean concrete.

CONSTRUCTION FOUNDATION DETAIL — POST-TENSIONED PODIUM SLAB ON DRILLED PIERS



PAGE 3: MEANS & METHODS, EQUIPMENT & SITE LOGISTICS

Equipment: Bauer BG 30 rotary drill rig, 42-inch auger with belling tool, 50-ton crane for rebar cages, Putzmeister 36m boom pump for podium pour (single 280 CY monolithic pour, night pour to control heat).

Sequence: Week 1: Mobilize, dewatering wells (2 wells, 20 gpm each, discharge to city storm with permit), excavation to -10 ft, proof roll. Week 2: Drill 12 piers (2 per day), integrity testing (PIT) on 100% piers. Week 3: Grade beam trenching, formwork, rebar, pour. Week 4: Podium formwork (Peri Skydeck), PT tendon install, MEP blockouts per BIM, pour and stressing.

Logistics: Limited laydown (40x60 ft), just-in-time rebar delivery (CMC Steel Houston), concrete from Martin Marietta Plant 12 (5-mile). Traffic control plan included for West Loop.

Safety: OSHA 10 for all workers, competent person for excavation, trench boxes for grade beams, PT stressing barricades, daily toolbox talks. EMR 0.71.

PAGE 4: QUALITY ASSURANCE, TESTING & CODE COMPLIANCE

Testing: Third-party testing by Terracon: slump, air, temp every truck, 6 cylinders per 50 CY (1,7,28 day), PT elongation records, 100% pier integrity (low-strain PIT), 2 static load tests not included (optional \$8,500 ea).

Compliance: 2021 IBC with Houston Amendments, ACI 318-19, PTI, ASTM. Wind: 115 mph ultimate. Seismic: SDC B. Geotech: Report by PSI Geotech 02/15/2026 (borings B1-B6). Allowance for 1 geotech site visit during pier drilling.

BIM: LOD 300 model for foundation/podium only, clash detection with MEP (plumbing sleeves for retail below). Navisworks federated model, 2 coordination meetings included.

Environmental: SWPPP compliance, dewatering permit from City of Houston, silt fence, wheel wash. SPCC plan for fuel.

PAGE 5: SCHEDULE, WARRANTY & SERVICE TERMS

Schedule: 28 calendar days from NTP to PT stressing completion, weather permitting (5 weather days included). Liquidated damages: \$500/day after Day 35, capped at 10% contract value. Early completion bonus: \$250/day up to \$5k.

Warranty: 2-year workmanship warranty per Texas Property Code, 1-year concrete warranty against structural defects (not cosmetic cracks <0.02 inch). PT warranty 10-year tendon per supplier (CMC). No warranty for dewatering-related settlement if dewatering stopped by owner.

Insurance: GL \$2M/\$4M, Auto \$1M, Umbrella \$5M, Workers Comp statutory, Professional liability for PT design \$1M.

Exclusions: Mass excavation beyond -10 ft, soil disposal (classified as non-hazardous, but owner responsible for haul-off if contaminated), permanent dewatering system, elevator pit (separate package), lightning protection, structural steel above podium, fireproofing.

PAGE 6: PRICING, BREAKDOWN & FINANCIALS

Item	Scope Detail	Cost
Drilled Piers (12)	36-inch shaft, 60-inch bell, 40ft avg, rebar cage, concrete, PT	\$38,500
Grade Beams	30x36 inch, 380 LF, rebar, formwork	\$12,800
Podium PT Slab	12-inch, 6200 sqft, SCC 6000psi, PT tendons (1.2 tons), rebar	\$32,500
Waterproofing	Blueskin WP200 6200 sqft + protection + drainage + subdrain	\$8,200
Dewatering (temp)	2 wells, 3 weeks, permit, discharge	\$3,200
BIM & Engineering	LOD300, 2 meetings, SER allowance \$1,500	\$2,400
Mobilization	Rig, crane, pump, cleanup	\$4,150
SUBTOTAL	Labor + Materials	\$101,750
Less Early Pay Discount	2% if paid within 10 days	-\$3,000
TOTAL	Net, ex-tax	\$98,750
Add-On	Static load test (each)	\$8,500 optional

Payment Terms: Net 20, 30% mobilization, 40% after piers, 30% after podium stressing. Retainage 10% until acceptance, 5% after 30 days. Bond: Performance & Payment bond available at 2.5% additional. Price lock 60 days, then ACI materials index escalation (cement + steel).

Value Engineering: If owner allows straight shaft (no bell) with deeper embedment 48ft, deduct \$4,200 but requires SER approval and geotech re-analysis.

Notes: This is a modest \$100k slice of \$140M total project – Atlas views this as gateway to upper levels.